

Week 2: Forward Markets and Parity Conditions

FINA3020 at CUHK Business School

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- Forward markets and forward premiums
- Covered interest parity (CIP), executable quotes, and order-book arbitrage
- CIP funding desk; post-2008 deviations and Hong Kong interest rates
- Uncovered interest parity (UIP) and the forward discount puzzle
- Purchasing power parity (PPP), real exchange rates, and real interest parity

A forward exchange contract is a bilateral agreement to buy or sell currency at an agreed-upon rate [today] to be transacted on an agreed-upon future date.

- The **value date** is the time when the exchange of currencies occurs for a deliverable forward
 - A non-deliverable forward (NDF) has net [local] cash settlement instead
- The **forward rate** $F_{t,T}$ is set at inception time t for settlement at time T
- A forward is usually a customized OTC contract between a dealer and a client
- A firm can use a forward to fix the home-currency value of an upcoming foreign cash flow

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Consider these fictional quotations for USD/HKD from Hong Kong's perspective across maturities:

Tenor / Value Date	Dealer Bid ($\frac{\text{HKD}}{\text{USD}}$)	Dealer Ask ($\frac{\text{HKD}}{\text{USD}}$)	Spread (pips)
Spot ($T = 0$)	7.8000	7.8002	2.0
6-Month Forward ($T = 180\text{d}$)	7.7970	7.7974	4.0
12-Month Forward ($T = 360\text{d}$)	7.7930	7.7936	6.0

- **Quote Level:** Forwards trade below spot ($F < S$). The 6-month bid is 7.7970 vs spot 7.8000. The USD buys fewer HKD forward, trading at a **forward discount**.
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In interbank FX markets, dealers quote **forward points** (swap points, pips) relative to spot:

$$\text{Outright Forward Rate } F_{t,T} = S_t + \frac{\text{Forward Points}}{\text{Pip Multiplier}} \quad (\text{remember 1 pip} = 0.0001, \text{ or } 0.01 \text{ for JPY})$$

Outright bid is *always* below outright ask ($F^{\text{bid}} < F^{\text{ask}}$), so order books never cross.

Forward Discount (Points Subtracted)

- 6-month points quote: $-30/-28$ pips
- $F^{\text{bid}} = S^{\text{bid}} - 0.0030 = 7.8000 - 0.0030 = 7.7970$
- $F^{\text{ask}} = S^{\text{ask}} - 0.0028 = 7.8002 - 0.0028 = 7.7974$
- Voice shorthand “30/28”: subtracting larger from bid preserves $F^{\text{bid}} < F^{\text{ask}}$.
- USD buys fewer HKD forward: **USD forward discount**.

Forward Premium (Points Added)

- Points quote: $+25/+30$ pips
- $F^{\text{bid}} = S^{\text{bid}} + 0.0025$
- $F^{\text{ask}} = S^{\text{ask}} + 0.0030$
- Base currency buys more quote currency forward than spot ($F > S$).

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The simple annualized forward premium or discount scales the percentage difference between the forward and spot rates:

$$p_{F,\text{annualized}} = \frac{F_{t,T} - S_t}{S_t} \times \frac{\text{days per year}}{T} = \frac{\text{Forward Points}}{S_t \times \text{Multiplier}} \times \frac{\text{days per year}}{T}$$

where $S_t \equiv S_{F/H,t}$ is the direct quote (home per foreign) and T is maturity in days.

Annualized discount for our 6-month USD/HKD example ($S = 7.8000$, $F = 7.7970$, $T = 180$ d):

$$p_{F,\text{USD}} = \frac{7.7970 - 7.8000}{7.8000} \times \frac{360}{180} = \frac{-0.0030}{7.8000} \times 2 = -0.0769\% \text{ per year } (-7.7 \text{ bps}).$$

Foreign-currency discount

$$F_{t,T} < S_t \implies p_F < 0$$

Forward points < 0 . Foreign currency buys fewer home currency units forward than spot.

Foreign-currency premium

$$F_{t,T} > S_t \implies p_F > 0$$

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Two Sides of the Same Trade (Close Approximation)

Because $p_{F,\text{HKD}} = -\frac{F-S}{F} \cdot \frac{360}{T} = -p_{F,\text{USD}} \cdot \frac{S}{F} \approx -p_{F,\text{USD}}$, a base currency forward discount (-0.0769%) mirrors a quote currency forward premium ($+0.0770\%$). The tiny wedge is $\frac{S}{F} = 1.000385$.

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The **forward curve** plots outright forward rates $F(T)$ or forward points across maturities: Overnight (ON), 1W, 1M, 2M, 3M, 6M, 12M (1Y).

Upward-Sloping (Premium)

- Quote rate $>$ Base rate ($r_q > r_b$)
- Points > 0 and grow with maturity T
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Downward-Sloping (Discount)

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Around 80% of international trade volume uses trade credit or other deferred payment. Example: I agree today to 1) purchase 900 German cars that will arrive at HK's port in 1 month and 2) send the payment of 36 million EUR in 3 months

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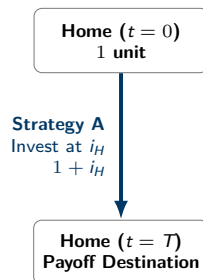
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CIP connects spot rates, forward rates, and interest rates in two currencies. Compare two investment strategies for 1 unit of domestic currency over horizon T :

Strategy A (Domestic Route)

$$\text{Payoff}_A = 1 + i_H$$



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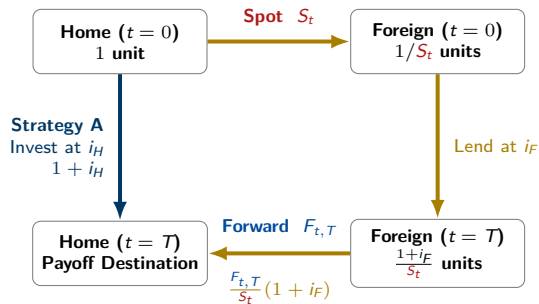
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Strategy B (Covered Foreign Route)

- Convert spot: receive $1/S_t$ foreign
- Invest abroad: earn foreign rate i_F
- Sell forward at $F_{t,T}$ to lock in:

$$\text{Payoff}_B = \frac{F_{t,T}}{S_t}(1 + i_F)$$



No Arbitrage (CIP): $\text{Payoff}_A = \text{Payoff}_B$

With matched maturities and no transaction costs, credit risk, balance-sheet costs, or capital controls, the two strategies must have the same return:

$$1 + i_H = \frac{F_{t,T}}{S_t}(1 + i_F)$$

Rearrange the equation to find the no-arbitrage forward rate:

$$F_{t,T} = S_t \cdot \frac{1 + i_H}{1 + i_F}$$

- If $i_H > i_F$, then $F_{t,T} > S_t$: the foreign currency is at a forward premium, so the higher-interest home currency is at a forward discount.
- If $i_H < i_F$, then $F_{t,T} < S_t$: the foreign currency is at a forward discount, so the lower-interest home currency is at a forward premium.

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Taking natural logarithms of the exact CIP condition $1 + i_H = \frac{F_{t,T}}{S_t}(1 + i_F)$ gives:

$$\ln(1 + i_H) = \ln\left(\frac{F_{t,T}}{S_t}\right) + \ln(1 + i_F)$$

Using first-order Taylor approximations $\ln(1 + x) \approx x$ and $\ln(F/S) \approx \frac{F-S}{S}$:

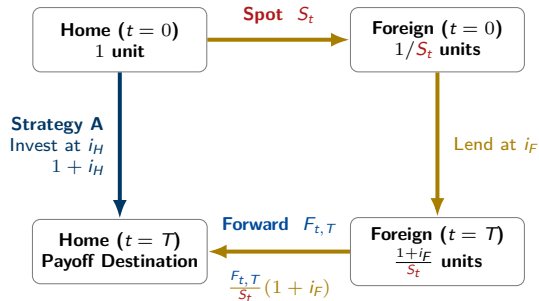
$$\frac{F_{t,T} - S_t}{S_t} \approx i_H - i_F$$

Pricing Intuition

The forward premium on the foreign currency is approximately the home interest rate minus the foreign interest rate, provided rates and the forward premium are modest and measured over the same horizon.

Assume $S_{\text{EUR/USD}} = 1.2000$ USD per EUR,
 $r_{\text{USD}} = 4\%$ and $r_{\text{EUR}} = 2\%$.

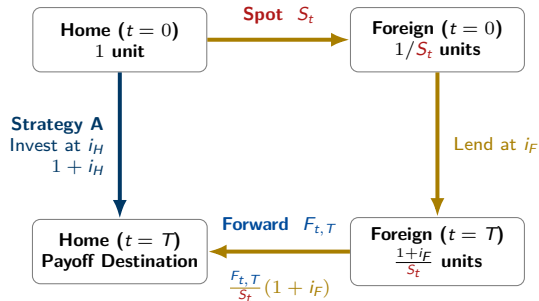
- ① Convert each annual rate into a 90-day holding-period return.
- ② Compare a USD deposit with a covered EUR deposit. What forward rate makes their returns equal?
- ③ Calculate the EUR forward premium over 90 days and annualize it.



Individual work: 3 minutes. Write currency units beside each cash flow.

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The holding-period returns are $i_{\text{USD}} = 0.04(90/360) = 0.01$ and $i_{\text{EUR}} = 0.02(90/360) = 0.005$.

$$1.01 = \frac{F}{1.2000}(1.005) \implies F = 1.2000 \frac{1.01}{1.005} = 1.205970$$

$$\text{90-day premium} = \frac{1.205970 - 1.2000}{1.2000} = 0.4975\%$$

$$\text{Annualized premium} = 0.4975\% \times 4 = 1.9900\%$$

How do interest rates shape the forward curve?

Scan the QR code and enter your details. Work and submit individually, then discuss your reasoning.

Three challenges

- ① **Dealer quotes:** Convert forward points into an outright rate and identify the premium or discount.
- ② **Corporate hedging:** Calculate the euros needed to cover a future dollar payment.
- ③ **Rate shock:** Predict how a change in interest rates shifts the forward curve.

Compare USD/HKD, EUR/USD, and USD/JPY using the **Outright Rates** and **Forward Points** views.



[Launch Forward Curve Game](#)

Exit Reflection

With fixed, unequal rates, why do longer maturities have larger absolute forward points? Submit one sentence.

What happens when CIP does not hold? Suppose $1 + i_H < \frac{F_{t,T}}{S_t}(1 + i_F)$.

An arbitrageur can use the following trade:

- ① Borrow N units of domestic currency at interest rate i_H .
- ② Convert N domestic currency to foreign currency at spot rate S_t : Receive N/S_t foreign units.
- ③ Lend N/S_t foreign units in foreign money market at interest rate i_F : Receive $\frac{N}{S_t}(1 + i_F)$ at maturity.
- ④ Sell foreign proceeds forward at rate $F_{t,T}$: Lock in domestic cash return of $\frac{N}{S_t}(1 + i_F)F_{t,T}$.
- ⑤ Repay domestic borrowing $N(1 + i_H)$.

The arbitrage profit is $N \left[\frac{F_{t,T}}{S_t}(1 + i_F) - (1 + i_H) \right]$.

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- ⑤ Repay domestic borrowing $N(1 + i_H)$.

The arbitrage profit is $N \left[\frac{F_{t,T}}{S_t}(1 + i_F) - (1 + i_H) \right]$.

What happens when CIP does not hold? Suppose $1 + i_H < \frac{F_{t,T}}{S_t}(1 + i_F)$.

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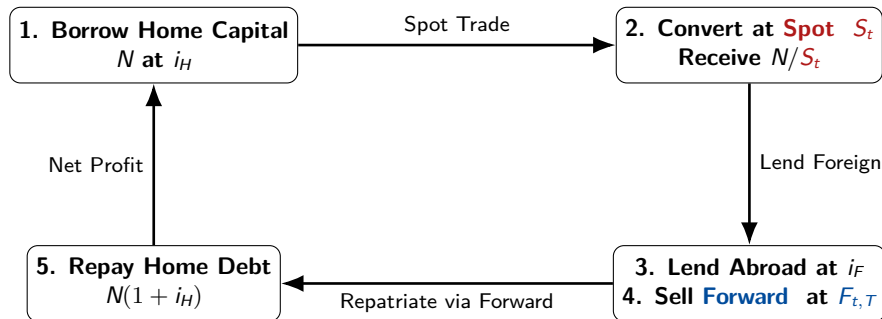
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These trades change the spot rate, forward rate, and interest rates until the profit disappears.

Market Quotes & Rates:

- **Spot Rate:** $S_{\text{GBP/USD},t} = 1.3000 \frac{\text{USD}}{\text{GBP}}$
- **1-Year Forward:** $F_{\text{GBP/USD},t,1} = 1.3200 \frac{\text{USD}}{\text{GBP}}$
- **US Interest Rate:** $i_{\text{USD}} = 2.0\%$
- **UK Interest Rate:** $i_{\text{GBP}} = 1.0\%$

Return Comparison:

$$\text{Covered GBP return} = \frac{1.3200}{1.3000}(1.01) - 1 = 2.5538\% > 2.0000\% = \text{USD return}$$

Arbitrage Result: Under the assumptions of zero cost, and simultaneous execution, borrowing \$1M USD to invest in covered GBP yields +\$5,538.46 USD.

Keep $S_{\text{GBP/USD}} = 1.3000$, $i_{\text{USD}} = 2\%$, and $i_{\text{GBP}} = 1\%$. Now let the one-year forward be $F = 1.3000$.

$$F^{\text{CIP}} = 1.3000 \frac{1.02}{1.01} = 1.312871 > F$$

Today: borrow GBP, lend USD	At maturity
Borrow GBP 1,000; sell spot for USD 1,300	USD deposit pays 1,326
Invest USD 1,300 at 2%	Buy GBP 1,010 forward for USD 1,313
Buy GBP 1,010 forward today	Repay GBP debt; retain USD 13

Why is the forward purchase GBP 1,010 rather than GBP 1,000?

Bid-ask spreads, borrowing-lending wedges, and intermediary balance-sheet costs create a band where arbitrage is unprofitable:

- **Bid-Ask Spreads:** $S_{\text{bid}} < S_{\text{ask}}$ and $F_{\text{bid}} < F_{\text{ask}}$
- **Interest Rate Spreads:** $i^{\text{borrow}} > i^{\text{lend}}$ in both currencies

Core Intuition: Covered interest arbitrage is only executed if the return differential exceeds total transaction costs.

Takeaway: Even if you calculate a CIP deviation, you might not be able to achieve risk-free profit after considering executable quotes, funding costs, credit terms, capital charges, and balance-sheet constraints.

Bid-Ask No-Arbitrage Inequalities →

Rates below match the forward's maturity; S and F are home currency per foreign unit.

Borrow / invest	Spot execution	Forward execution
Borrow home / lend foreign	Buy foreign at S_{ask}	Sell foreign at F_{bid}
Borrow foreign / lend home	Sell foreign at S_{bid}	Buy foreign at F_{ask}

A profitable trade requires one of these inequalities:

$$(1 + i_F^{\text{lend}})F_{\text{bid}} > (1 + i_H^{\text{borrow}})S_{\text{ask}}$$

$$(1 + i_H^{\text{lend}})S_{\text{bid}} > (1 + i_F^{\text{borrow}})F_{\text{ask}}$$

Think: Can a deviation calculated with mid-quotes survive neither test? Explain which prices an arbitrageur actually receives.

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Think: Can a deviation calculated with mid-quotes survive neither test? Explain which prices an arbitrageur actually receives.

One-year AUD/HKD quotes, in HKD per AUD. Borrow and lend at $i_{\text{HKD}} = 4\%$, $i_{\text{AUD}} = 1\%$; no other costs. At the top, spot bid/ask is 5.990/6.000 and forward bid/ask is 6.200/6.210.

Level	Spot asks: buy AUD today		Forward bids: sell AUD at T	
	Price	AUD size	Price	AUD delivery size
1	6.000	800	6.200	606
2	6.010	700	6.190	909
3	6.020	1,000	6.180	1,010

① Compare F^{CIP} at the spot midpoint with the forward midpoint.

② Now use executable quotes to select the trade. Which best-quote leg limits its size?

Work with a neighbour. Assume simultaneous fills, divisible quantities, and fixed books.

Walking exercise

The midpoint benchmark is $5.995(1.04/1.01) = 6.17307 < 6.205$.

Borrow HKD, buy AUD spot, deposit AUD, and sell its maturity proceeds forward.

q AUD today $\longrightarrow$ $1.01q$ AUD at maturity

$$q \leq 800, \quad 1.01q \leq 606 \quad \implies \quad q_{\max} = \min(800, 606/1.01) = 600$$

Cash flow	Amount
HKD borrowed today: $600(6.000)$	3,600.00
HKD debt at maturity: $3,600(1.04)$	3,744.00
HKD received forward: $606(6.200)$	3,757.20
Profit at maturity	HKD 13.20

The forward leg binds. Matching both legs at 600 AUD would leave the interest unhedged.

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After the first trade, 200 AUD remain at the spot ask of 6.000
The next forward bid is 6.190 for 909 AUD delivered.

$$\text{Profit per AUD bought spot} = m(a, f) = 1.01f - 1.04a$$

$$\Delta q = \min \left(\text{remaining spot size}, \frac{\text{remaining forward delivery size}}{1.01} \right)$$

- ① At each pair of prices, compute $m(a, f)$ before taking a new tranche.
- ② Reduce spot depth by Δq and forward depth by $1.01\Delta q$.
- ③ Move down whichever book runs out. Stop when marginal profit is non-positive.

Pair work: 3 minutes. Find total spot purchases and total HKD profit. Do not round intermediate quantities.

[Order books](#)

Spot ask	Forward bid	Spot AUD	Forward AUD	HKD/AUD margin	HKD profit
6.000	6.200	600	606	0.0220	13.20
6.000	6.190	200	202	0.0119	2.38
6.010	6.190	700	707	0.0015	1.05
Total		1,500	1,515		16.63

Next pair: $1.01(6.180) - 1.04(6.020) = -0.0190$ HKD per AUD. Stop.

- Both prices worsen as we walk the books: all later margins are also negative.
- The first tranche earns the most per AUD. Check if later tranches have *positive marginal profit*.
- With uncrossed books and these common borrowing/lending rates, the reverse trade is already unprofitable at the best quotes and worsens with depth.

Keep the best quotes $S_{\text{ask}} = 6.000$ and $F_{\text{bid}} = 6.200$. Let the HKD borrowing rate rise from 4% to $4\% + x$; AUD lending stays at 1%.

$$1.01(6.200) - (1.04 + x)(6.000) = 0$$

$$x^* = \frac{1.01(6.200)}{6.000} - 1.04 = 0.0036667$$

A Spread Is a Difference in Rates

The trade breaks even at an additional borrowing spread of **0.3667 percentage points**, or **36.67 basis points**.

At a one-percentage-point spread, the margin is $1.01(6.200) - 1.05(6.000) = -0.038$ HKD per AUD. **Choose no trade.** Worse book prices cannot restore profitability.

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10 minutes. Work on your own device, but you can discuss with classmates.

- 1 Choose **borrow HKD**, **borrow USD**, or **no trade**
- 2 Write the cash flow at $t = 0$ and repayment at T
- 3 Scan QR code or open [CIP Funding Desk Game](#)
- 4 Increase the spread until no-arbitrage bounds hold
- 5 Submit individual exit response explaining your choice

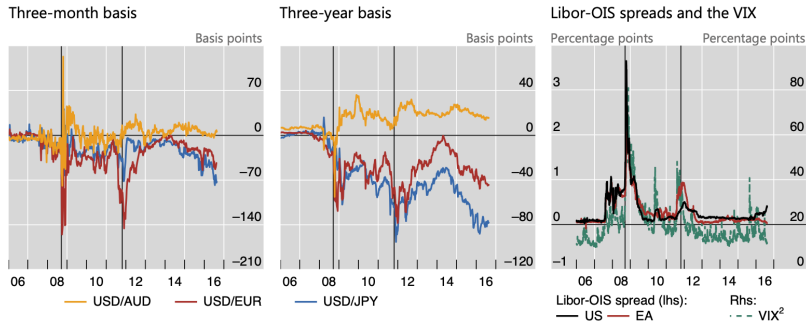
Exit Reflection

“Which executed quote and funding rate determine each no-arbitrage bound?”



[Launch CIP Desk](#)

Borrowing USD *synthetically* through FX swaps has become relatively expensive.



¹ The vertical lines indicate 15 September 2008 (Lehman Brothers file for Chapter 11 bankruptcy protection) and 26 October 2011 (euro area authorities agree on debt relief for Greece, leveraging of the European Financial Stability Facility and the recapitalisation of banks). ² Chicago Board Options Exchange S&P 500 implied volatility index; standard deviation, in percentage points per annum.

Source: Borio, McCauley, McGuire, Sushko (2016), Graph 1; historical data through 2016

Read the figure: Did the basis return to zero when the stress indicators fell? Under this figure's convention, a negative basis = CIP deviation with premium for synthetic USD borrowing.

An FX swap combines a spot exchange with a reverse exchange at a fixed forward rate. Use the foreign borrowing rate i_F . For S and F in USD per foreign unit:

$$1 + i_{\text{USD}}^{\text{synthetic}} = \frac{F}{S}(1 + i_F)$$

- Borrow foreign currency, sell it spot for USD, and buy the foreign repayment forward.
- Example: $S = 1.20$, $F = 1.236$, $i_F = 2\%$ for one year. Synthetic USD funding costs $(1.236/1.20)(1.02) - 1 = 5.06\%$.
- If direct USD borrowing costs 4.80%, synthetic funding is **26 basis points more expensive** before further execution costs.

Sign Convention

Here we report synthetic minus direct USD cost. A positive cost premium corresponds to a negative foreign-leg basis under the preceding figure's convention.

- ① Firms and investors demand FX hedges, creating persistent pressure on forward prices.
- ② A bank taking the opposite position uses funding, credit limits, collateral, and balance-sheet capacity even when its currency exposure is hedged.
- ③ Leverage requirements and internal risk limits make that capacity costly. The bank expands only if the return covers its marginal costs.

Example: A 26 bp gross spread is unattractive if additional execution and balance-sheet charges total 30 bp. Increasing volume also uses more scarce capacity.

Think: Why might the gap persist outside a financial crisis?

Source: Borio et al. (2016). Regulation is one channel; hedging demand and internal constraints also matter. Week 8 develops the funding institutions.

Under the Linked Exchange Rate System, HKMA convertibility undertakings bound USD/HKD at 7.75 and 7.85 for licensed banks.

Pressure	HKMA transaction	Adjustment in HKD money markets
Weak HKD: 7.85	Buys HKD, sells USD	Aggregate Balance contracts; upward pressure on HKD rates
Strong HKD: 7.75	Sells HKD, buys USD	Aggregate Balance expands; downward pressure on HKD rates

The Aggregate Balance is banks' clearing balances at the HKMA.

Implication: Open capital mobility and the exchange-rate commitment limit independent monetary policy. HKD rates track USD rates over time, with local liquidity affecting the spread.

Source: HKMA (2018), Linked Exchange Rate System operations. Full regime analysis: Week 3.

CIP for USD/HKD, with simple annual rates and year fraction τ :

$$\frac{F}{S} = \frac{1 + r_{\text{HKD}}\tau}{1 + r_{\text{USD}}\tau}$$

Hypothetical 90-day example: $S = 7.8000$, $r_{\text{HKD}} = 3\%$, $r_{\text{USD}} = 5\%$, and $\tau = 90/360$ for both rates.

$$F = 7.8000 \frac{1.0075}{1.0125} = 7.76148$$

- A small forward discount can offset a material *annualized* interest gap over a short horizon.
- Exact equality of rates follows from $F = S$ under frictionless CIP; the actual peg has a band and changing liquidity conditions.

Why does borrowing cheaper HKD to hold unhedged USD still involve risk?

Source: Mechanism: HKMA (2018). Hypothetical rates; no forward-rate guarantee from HKMA.

CIP uses a forward contract to fix the future exchange rate. UIP considers an investor who does not use a forward contract and therefore bears exchange rate risk. Under risk neutrality, compare expected returns:

$$1 + i_H = \frac{\mathbb{E}_t[S_{t+T}]}{S_t}(1 + i_F)$$

where $\mathbb{E}_t[S_{t+T}]$ is the expected spot rate at maturity T .

Linear approximation:

$$\mathbb{E}_t \left[\frac{\Delta S_{t+T}}{S_t} \right] \approx i_H - i_F$$

With S_t measured in home currency per unit of foreign currency, the expected depreciation of the home currency equals $i_H - i_F$.

Expected Return, Not Arbitrage

UIP is an equilibrium benchmark under risk neutrality. Unlike CIP, it cannot be directly arbitrated because S_{t+T} is unknown.

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UIP is an equilibrium benchmark under risk neutrality. Unlike CIP, it cannot be directly arbitrated because S_{t+T} is unknown.

CIP: hedged and arbitrage-based

$$\frac{F_{t,T} - S_t}{S_t} \approx i_H - i_F$$

The forward contract removes exchange-rate risk.

UIP: unhedged and expectation-based

$$\mathbb{E}_t \left[\frac{\Delta S_{t+T}}{S_t} \right] \approx i_H - i_F$$

The future spot rate remains uncertain.

Unbiasedness Hypothesis

Combining CIP with risk-neutral UIP implies

$$F_{t,T} = \mathbb{E}_t[S_{t+T}]$$

but this is not a no-arbitrage result by itself.

One common simple-return version of the Fama (1984) regression is:

$$\frac{S_{t+T} - S_t}{S_t} = \alpha + \beta \left(\frac{F_{t,T} - S_t}{S_t} \right) + \varepsilon_{t+T}$$

- CIP plus UIP and rational expectations predict $\alpha = 0$ and $\beta = 1$.
- Many historical short-horizon samples give $\beta < 0$; the result varies across currencies and periods.
- This is the **forward discount puzzle**: high-interest currencies often appreciate instead of depreciating.

Example: Each observation covers the same one-year horizon. Let $x = (F - S)/S$ and $y = (S_T - S)/S$, both expressed in percentage points.

Observation	Forward premium x	Realized spot change y
A	-2	+1
B	0	0
C	+2	-1

- 1 Find mean $\bar{x}$ (for CIP), mean $\bar{y}$ (for UIP), and slope $\hat{\beta} = \frac{\sum(x-\bar{x})(y-\bar{y})}{\sum(x-\bar{x})^2}$.
- 2 Under CIP, which currency has the higher interest rate in observation C?
- 3 Does the realized exchange-rate movement in C match the UIP prediction?

$$\bar{x} = \bar{y} = 0, \quad \hat{\beta} = \frac{(-2)(1) + (2)(-1)}{(-2)^2 + 2^2} = -0.5, \quad \hat{\alpha} = 0$$

In observation C, $x = +2\%$ implies a higher *home* interest rate under CIP.

UIP predicts home depreciation; $y = -1\%$ instead records home appreciation.

- A negative slope reverses the predicted relationship in this sample.
- Three observations do not establish a population result. Real studies need uncertainty estimates and a clear sample and horizon.
- A profitable average carry return does not establish a riskless arbitrage.

Source: For variation in actual estimates across periods, see Bussière, Chinn, Ferrara, Heipertz, The New Fama Puzzle (2022 revision).

In frictionless, competitive markets, the law of one price says an identical tradable good should have the same price when expressed in a common currency

- Necessary: same place, time, quality, and contract terms-

Without trade costs:

$$P_{i,H} = S_{F/H} \cdot P_{i,F}$$

where $P_{i,H}$ is the home price of good i , $P_{i,F}$ is its foreign price, and $S_{F/H}$ is home currency per unit of foreign currency.

If $P_{i,H} < S_{F/H} \cdot P_{i,F}$, trade arbitrageurs buy good i in the home market and sell it in the foreign market until prices align.

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The same sweater (same brand, size, etc) is priced at HKD 200 in Hong Kong and USD 30 in the US. The exchange rate is 7.8 HKD per USD.

$$\text{Hong Kong purchase cost in USD} = 200/7.8 = 25.64$$

$$\text{Gross margin} = 30(7.8) - 200 = \text{HKD } 34 \text{ per sweater}$$

- ① Where should a trader buy and sell? What happens to prices as trade expands?
- ② If shipping, tariffs, and handling cost HKD 40 per sweater, is the trade profitable?
- ③ Would an identical argument work for a haircut or an apartment?

Assume a fixed conversion rate and a guaranteed sale for the calculation. Actual goods trade also involves timing, quality, and demand risk.

Absolute PPP extends the law of one price from one good to the same representative consumption basket in two countries. Absolute PPP:

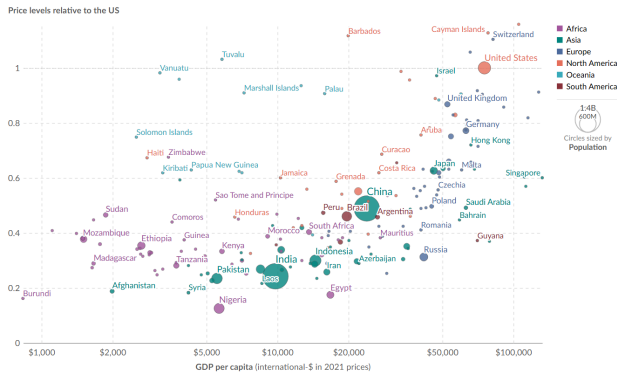
$$S_{F/H}^{\text{PPP}} = \frac{P_H}{P_F}$$

where P_H is domestic price level of consumption basket and P_F is foreign price level. The nominal exchange rate equals the ratio of the two price levels.

Is this realistic?

- Haircuts, housing, and many health services cannot be shipped across borders.
- Shipping costs, tariffs, and non-tariff barriers create a band within which goods arbitrage is not profitable.
- Wages and prices adjust slowly.
- Firms can set different prices in separated markets.
- National price indexes use different baskets, quality adjustments, and expenditure weights.

Lower-income countries tend to have lower price levels than wealthier countries.



Data source: World Development Indicators - World Bank (2025); Eurostat, OECD, IMF, and World Bank (2025) - [Learn more about this data](#)

Price Level vs. Real GDP per Capita

Source: Penn World Table / World Bank

The Balassa-Samuelson mechanism:

- Rich countries tend to have especially high productivity in tradable sectors relative to nontradable sectors.
- Labor mobility transmits higher tradable-sector wages to nontradable sectors.
- With weaker productivity gains in nontradables, their prices—and the aggregate price level—are higher.

Use a common currency. A tradable good sells for 10 in both countries. Competitive firms hire labour; workers can move between sectors within each country.

	Country A	Country B
Tradable output per hour	1	2
Hourly wage: price $\times$ productivity	10	20
Haircuts per hour	1	1
Haircut price needed to cover wages	10	20

A basket containing one tradable and one haircut costs 20 in A and 30 in B.

Discussion: Why does higher productivity in B's tradable sector raise the price of haircuts even though haircut productivity is unchanged?

The real exchange rate $Q_{F/H,t}$ is the price of the foreign basket relative to the home basket:

$$Q_{F/H,t} = \frac{S_{F/H,t} \cdot P_{F,t}}{P_{H,t}}$$

Interpretation:

- If $Q_{F/H,t}$ rises, the home currency depreciates in real terms. The home basket becomes cheaper relative to the foreign basket.
- If $Q_{F/H,t}$ falls, the home currency appreciates in real terms. The home basket becomes more expensive.
- Absolute PPP implies $Q_{F/H,t} = 1$.

Home is the US. Foreign is the UK.

A comparable basket costs USD 150 in the US and GBP 100 in the UK.

The spot quote is $S_{\text{GBP/USD}} = 1.3500$ USD per GBP.

- ① Express the UK basket price in USD, then calculate $Q = SP_{\text{UK}}/P_{\text{US}}$.
- ② Calculate the absolute-PPP exchange rate. Which currency is overvalued relative to that benchmark?
- ③ If only the nominal exchange rate adjusts to absolute PPP, what percentage change in S is required?

Use prices of comparable baskets. In reality, separately rebased national CPI indexes do not identify absolute price-level differences.

$$P_{UK} \text{ in USD} = 1.35(100) = 135, \quad Q = 135/150 = 0.90$$

The home basket is more expensive: the USD is overvalued *relative to absolute PPP*.

$$S^{PPP} = 150/100 = 1.50 \text{ USD/GBP}, \quad \frac{1.50}{1.35} - 1 = 11.11\%$$

- The GBP price in USD must rise 11.11% to reach this PPP benchmark if basket prices stay fixed.
- In the inverse quote, the USD's GBP value falls 10%. Percentages depend on the denominator.

Interpretation

$Q < 1$ compares basket prices. It does not guarantee a future correction or a profitable currency trade.

Relative PPP focuses on changes in exchange rates and inflation differentials over time. Relative PPP:

$$\frac{S_{F/H,t+1} - S_{F/H,t}}{S_{F/H,t}} \approx \pi_{H,t+1} - \pi_{F,t+1}$$

where π_H is home inflation rate and π_F is foreign inflation rate. If home inflation is higher, the home currency should depreciate over a long horizon by approximately the inflation differential.

From $Q = SP_F/P_H$, the exact one-period change is

$$\frac{Q_1}{Q_0} = \frac{S_1}{S_0} \frac{1 + \pi_F}{1 + \pi_H}$$

Constant Q: If US inflation is 4% and UK inflation is 2%,

$$\frac{S_1}{S_0} = \frac{1.04}{1.02} = 1.019608$$

The GBP price in USD rises 1.96%, approximately the 2 percentage point inflation gap.

Changing Q: Suppose instead S rises 5%, UK inflation is 5%, and US inflation is 2%:

$$\frac{Q_1}{Q_0} = 1.05 \frac{1.05}{1.02} = 1.080882$$

Real home depreciation is 8.09%, approximately $5 + 5 - 2 = 8\%$. Relative PPP would require Q to stay constant, which it does not in this second case.

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Under risk neutrality, rational expectations, and ex ante relative PPP, combine the Fisher effect, UIP, and expected inflation:

- Nominal Fisher Effect: $i_H \approx r_H^e + \pi_H^e$ and $i_F \approx r_F^e + \pi_F^e$.
- Uncovered Interest Parity: $\mathbb{E}[\Delta s] \approx i_H - i_F$.
- Relative PPP: $\mathbb{E}[\Delta s] \approx \pi_H^e - \pi_F^e$.

Combining these equations yields Real Interest Parity (RIP):

$$r_H^e = r_F^e$$

Real expected interest rates equalize only under these strong joint assumptions.

Consider these one-year rates: $i_{US} = 5\%$, $i_{UK} = 3\%$, $\pi_{US}^e = 2\%$, and $\pi_{UK}^e = 1\%$.

$$r_{US}^e \approx 5\% - 2\% = 3\%, \quad r_{UK}^e \approx 3\% - 1\% = 2\%$$

- UIP predicts approximately 2% nominal home depreciation.
- Ex ante relative PPP predicts approximately 1% nominal home depreciation.
- These predictions cannot both hold with the stated rates. If UIP holds, expected real home depreciation is approximately $2\% - 1\% = 1\%$.

Consistency Check

Equal expected real rates require the additional restriction of a constant expected real exchange rate. Fisher and UIP alone do not deliver that restriction.

For each claim, write “true” or “false” and one sentence explaining your answer.

- ① A currency at a forward premium will appreciate in the spot market.
- ② Buying 1,000 foreign units spot requires selling 1,000 units forward to hedge a deposit earning 2%.
- ③ A credible peg makes all home and anchor-currency interest rates identical.
- ④ A real exchange rate of 0.90 implies that the home currency will depreciate by 10% next year.

- ① **False.** The forward rate prices a contract. An expected future spot rate requires additional assumptions.
- ② **False.** Sell 1,020 foreign units forward to cover principal and interest at the deposit's maturity.
- ③ **False.** Rates are linked, but forward points, liquidity, credit terms, and the maturity can support differences.
- ④ **False.** $Q = 0.90$ describes a relative basket price; it does not identify the timing or size of a future nominal movement.

Condition	Equation	Reason or assumption
Covered Interest Parity (CIP)	$\frac{F - S}{S} \approx i_H - i_F$	No-arbitrage benchmark
Uncovered Interest Parity (UIP)	$\mathbb{E} \left[\frac{\Delta S}{S} \right] \approx i_H - i_F$	Risk neutrality / expectations
Absolute PPP	$S = \frac{P_H}{P_F}$	Frictionless trade / LOP
Relative PPP	$\frac{\Delta S}{S} \approx \pi_H - \pi_F$	Long-run inflation adjustment
Real Interest Parity (RIP)	$r_H^e = r_F^e$	Fisher + UIP + ex ante PPP

Next week we move from parity benchmarks to exchange rate expectations, private-sector forecasting, carry trades, and policy regimes:

- **Dornbusch Overshooting:** Sticky goods prices force spot rates to jump on monetary surprises.
- **Carry Trades & Tail Risk:** Positive carry returns compensate for negative skewness and liquidity spirals (e.g., August 2024 JPY unwind).
- **Private-Sector Forecasting (Melvin et al. 2013):** Why quants care about cross-sectional ranking rather than RMSE; the 4 factor styles (Carry, Momentum, Value, Macro).
- **Order Flow & Interview Playbook:** Using customer order flow to forecast spot moves; the 4-pillar currency trade pitch for macro desk interviews.
- **Regimes & Interventions:** HKMA Currency Board mechanics and sterilized vs unsterilized operations.

Bridge to Week 3

Parity conditions define the theoretical benchmark. Private-sector factor models, liquidity spirals, and policy constraints explain how currencies trade in practice.

Appendix: Step-by-Step CIA Calculation

Starting by borrowing US\$1,000,000:

- ① **Borrow Domestic Capital:** Borrow \$1,000,000 USD at 2.0%.
Repayment obligation = \$1,020,000 USD.

- ② **Convert Spot:** Convert USD to GBP at $S_t = 1.3000$:

$$\frac{\$1,000,000}{1.3000} = 769,230.77 \text{ GBP.}$$

- ③ **Lend Abroad:** Lend GBP in UK money market at 1.0%:

$$769,230.77 \times 1.01 = 776,923.08 \text{ GBP.}$$

- ④ **Forward Hedge:** Sell GBP proceeds forward at $F_{t,1} = 1.3200$:

$$776,923.08 \times 1.3200 = \$1,025,538.46 \text{ USD.}$$

- ⑤ **Net Profit:** $\$1,025,538.46 - \$1,020,000 = +\$5,538.46 \text{ USD.}$

Appendix: Bid-Ask No-Arbitrage Bounds

To prevent profitable covered interest arbitrage in either direction, quotes must satisfy two execution bounds:

Bound 1 (Borrow Home, Lend Foreign):

$$\frac{F_{\text{bid}}}{S_{\text{ask}}} (1 + i_F^{\text{lend}}) \leq 1 + i_H^{\text{borrow}}$$

Bound 2 (Borrow Foreign, Lend Home):

$$\frac{F_{\text{ask}}}{S_{\text{bid}}} (1 + i_F^{\text{borrow}}) \geq 1 + i_H^{\text{lend}}$$

Inside this band, transaction costs eat up any apparent CIP profit margin.

Reserve Exercise: Reverse CIA with Depth

One-year GBP/USD executable quotes (USD per GBP):

	Price	GBP size
Best spot bid / ask	1.300 / 1.302	1,000 each
Best forward bid / ask	1.298 / 1.300	808 each

USD borrowing/lending rates: 3% / 2%. GBP borrowing/lending rates: 1% / 0.5%.

- ① Test both arbitrage directions using the correct funding rates.
- ② If one works, size the trade at the best quotes and calculate USD profit.
- ③ What limits the GBP principal? How does an additional 50 bp GBP borrowing cost change the result?

Reserve Solution: Reverse CIA with Depth

Borrow GBP, lend USD: $1.300(1.02) - 1.300(1.01) = 0.013$ USD per GBP borrowed. The opposite direction loses: $1.298(1.005) - 1.302(1.03) = -0.03657$ USD per GBP deposited.

$$q_{\text{GBP borrowed}} = \min(1,000, 808/1.01) = 800$$

Sell GBP 800 for USD 1,040; the USD deposit grows to 1,060.80. Buy GBP 808 forward for USD 1,050.40 and repay the GBP debt.

$$\text{Profit} = 1,060.80 - 1,050.40 = \text{USD } 10.40$$

With GBP borrowing at 1.5%, re-size to $808/1.015 = 796.0591$ GBP. The margin becomes $1.300(1.02 - 1.015) = 0.0065$ USD per GBP; profit is $796.0591(0.0065) = \text{USD } 5.17$.

Reserve Exercise: A Change in Forward Depth

Return to the AUD/HKD classroom order books: spot asks are 6.000 for 800 AUD, 6.010 for 700, and 6.020 for 1,000. Forward bids remain 6.200, 6.190, and 6.180. Now the best forward bid offers 1,010

AUD delivery rather than 606. The second level still offers 909 AUD delivery. Rates remain 4% HKD and 1% AUD for one year, with borrowing at deposit rates.

- ① Which leg binds on the first tranche?
- ② Walk the books until the next trade becomes unprofitable.
- ③ How do total spot purchases and total profit compare with the original example?

Reserve Solution: A Change in Forward Depth

Spot ask	Forward bid	Spot AUD	Forward AUD	HKD profit
6.000	6.200	800	808	17.60
6.010	6.200	200	202	2.32
6.010	6.190	500	505	0.75
Total		1,500	1,515	20.67

The spot leg binds first. At the next prices, $1.01(6.190) - 1.04(6.020) = -0.0089$ HKD per AUD. Stop. Spot volume stays at 1,500 AUD, but profit rises by $20.67 - 16.63 = \text{HKD } 4.04$. An extra 404 AUD of forward delivery receives 6.200 rather than 6.190. **Interpretation:** Greater depth can improve the execution price without changing the optimal total quantity.

Reserve Exercise: Nominal Stability and Real Appreciation

Home is Hong Kong; foreign is the US. The same basket costs HKD 780 in Hong Kong and USD 100 in the US at $S_0 = 7.80$ HKD per USD. Over the next year, Hong Kong basket inflation is 5%, US basket inflation is 2%, and the nominal rate stays at 7.80.

- 1 Compute Q_0 and Q_1 . Which currency appreciates in real terms?
- 2 What nominal exchange rate would keep Q constant?
- 3 Explain why a fixed nominal exchange rate does not guarantee stable international purchasing power.

Reserve Solution: Nominal Stability and Real Appreciation

$$Q_0 = \frac{7.8(100)}{780} = 1, \quad Q_1 = \frac{7.8(102)}{819} = 0.971429$$

The HKD appreciates in real terms: Q falls by 2.86%. Hong Kong's basket becomes more expensive relative to the US basket.

$$S_1^{\text{constant } Q} = \frac{819}{102} = 8.029412 \text{ HKD/USD}$$

- Constant Q would require the USD price in HKD to rise 2.94%.
- The actual nominal rate is fixed in this scenario; the inflation gap instead changes the real exchange rate.
- A nominal peg disciplines the exchange rate, while domestic prices can still move relative to foreign prices.